

Phase-Aware Conformal Risk Control for Selective Antimicrobial Susceptibility Display

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Abstract

We study selective display of antibiotic-specific susceptibility predictions as information accrues. On AMR-UTI, deterministic feature masks approximate Empiric, Gram, Culture, and Sensitivity states. A gated phase-aware controller reaches 27.4% held-out coverage at 0.105 aggregate unsafe-display risk; a pooled phase-unaware comparator reaches 20.5% at 0.111. Because the systems also differ in model granularity, calibration, threshold structure, and gating, this is a descriptive system comparison rather than a pure phase-awareness ablation.

Keywords: Conformal risk control, selective prediction, antimicrobial resistance, clinical decision support, Venn-ABERS.

1. Setting and Method

Empiric urinary-tract-infection (UTI) therapy precedes susceptibility results by 24–72 h. We represent progressive information by deterministic Empiric, Gram, Culture, and Sensitivity feature masks. A *display* exposes an antibiotic-specific susceptibility probability for clinician review, not a treatment recommendation; *unsafe-display risk* is the fraction of displayed predictions whose observed label is resistant.

For each antibiotic $a \in \{\text{NIT}, \text{SXT}, \text{CIP}, \text{LVX}\}$ and phase $k \in \{0, 1, 2, 3\}$, the primary system fits a phase-antibiotic-specific LightGBM, calibrates with Venn-ABERS (Vovk and Petej, 2014), and applies prespecified deterministic safety checks that can suppress display before thresholding. Thresholds use the shifted-loss CRC-style rule motivated by Angelopoulos et al. (2024):

$$L_i(\tau) = \mathbf{1}\{\hat{p}_k(x_i) \geq \tau\}(\mathbf{1}\{r_i = 1\} - \varepsilon), \quad \hat{\tau}_k = \inf \left\{ \tau : \frac{n}{n+1} R_n(\tau) + \frac{1}{n+1} \leq 0 \right\},$$

with $R_n(\tau) = n^{-1} \sum_i L_i(\tau)$. We use $\varepsilon = 0.11$ as the primary benchmark operating point and 10–12% to examine the local risk-coverage trade-off; these are methodological study targets, not clinically validated deployment thresholds. Venn-ABERS and threshold selection reuse the same calibration observations (no nested split/cross-fitting); no test labels enter either step. Test covariates enter only a prespecified drift diagnostic, which triggered no weighted-threshold replacement. A later aggregate risk-tightening step can alter the initial threshold ordering, so final per-antibiotic thresholds are not guaranteed monotone and no formal cross-phase dominance is claimed.

Phase-unaware comparator. A single pooled LightGBM is fitted to rows stacked across antibiotics/phases, using phase-specific missingness masks but no explicit phase/antibiotic indicator, one global Platt calibrator, one global shifted-loss threshold, and no hard gates. It is therefore a complete-system comparator, not a single-factor ablation.

2. Data

AMR-UTI (Oberst et al., 2020; Kanjilal et al., 2020; Goldberger et al., 2000) contains 116,902 uncomplicated-UTI specimens with four resistance labels. The frozen temporal split is 80,962/14,376/21,564 train/calibration/test (69.3/12.3/18.5%; specimen-level disjoint), and the masks expose 125/224/286/790 columns. Patient IDs are unavailable, so repeated specimens may cross splits; phases are feature-availability overlays rather than timestamps.

3. Results

Main held-out result.

System	Unsafe risk	Coverage
Primary gated phase-aware	0.1052	27.43%
Pooled phase-unaware comparator	0.111	20.5%

The primary controller displays 90,157 decisions (9,486 unsafe). A paired specimen bootstrap ($B = 1000$) gives $\Delta_{\text{cov}} = +6.9$ pp (95% CI $[+6.6, +7.3]$) and $\Delta R = -0.0055$ (95% CI $[-0.0098, -0.0005]$). Because the systems also differ in gating, model granularity, calibration, and thresholds, these contrasts are descriptive.

Gated-primary phase risks (antibiotics aggregated).

Phase	Empiric	Gram	Culture	Sensitivity
Risk	0.1165	0.1094	0.1021	0.1003

The phasewise sequence is descriptive, not a formal cross-phase guarantee. In a secondary ungated threshold-only diagnostic, 4/14 displaying antibiotic-phase cells exceeded 0.11: NIT/Culture (0.143), SXT/Sensitivity (0.113), CIP/Empiric (0.115), and LVX/Empiric (0.118); NIT/Empiric and SXT/Empiric displayed none (risk undefined). These values are not attributed to the gated Primary. Calibration-to-test score drift is small (max KS = 0.012, Wasserstein = 0.003). In a separate synthetic sanity study, progressive information raised coverage from about 0.75 to 0.90, while reversing the schedule increased unsafe risk.

4. Related Work and Limitations

Selective classification (Geifman and El-Yaniv, 2017) studies prediction with a reject option; CRC (Angelopoulos et al., 2024) controls monotone loss, and Barber et al. (2023) treats nonexchangeability. We combine selective display with Venn-ABERS probabilities (Vovk and Petej, 2014). **Limitations:** the comparator is not a single-factor ablation; calibration observations are reused for thresholding; final thresholds need not remain monotone; phases are overlays; patient overlap is possible; and timestamped external validation is deferred.

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References

- A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024.
- R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816–845, 2023. doi:10.1214/23-AOS2276.
- Y. Geifman and R. El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 4878–4887, 2017.
- A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000.
- S. Kanjilal, M. Oberst, S. Boominathan, H. Zhou, D. C. Hooper, and D. Sontag. A decision algorithm to promote outpatient antimicrobial stewardship for uncomplicated urinary tract infection. *Science Translational Medicine*, 12(568):eaay5067, 2020. doi:10.1126/scitranslmed.aay5067.
- M. Oberst, S. Boominathan, H. Zhou, S. Kanjilal, and D. Sontag. AMR-UTI: Antimicrobial resistance in urinary tract infections (version 1.0.0). PhysioNet, 2020. doi:10.13026/se6w-f455.
- V. Vovk and I. Petej. Venn–Abers predictors. In *Uncertainty in Artificial Intelligence (UAI)*, pages 829–838, 2014.